



Krugman • Wells • Olney

»» Monopoly

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EVERYBODY MUST GET STONES



Got stones?

De Beers Diamond Company

What you will learn in this chapter:

- ▶ The significance of **monopoly**, where a single **monopolist** is the only producer of a good
- ▶ How a monopolist determines its profit-maximizing output and price
- ▶ The difference between monopoly and perfect competition, and the effects of that difference on society's welfare
- ▶ How policy makers address the problems posed by monopoly
- ▶ What **price discrimination** is, and why it is so prevalent when producers have **market power**

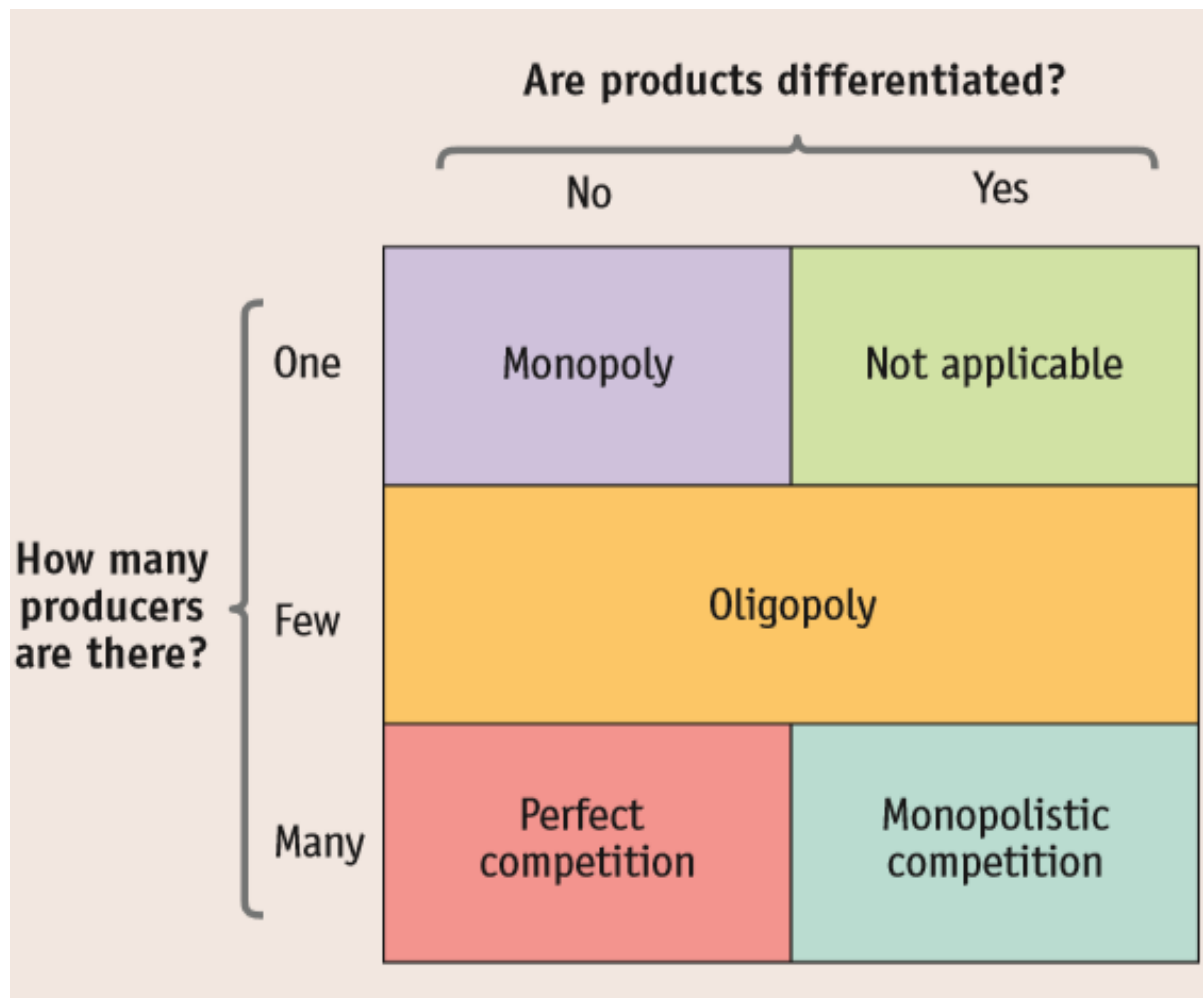
Types of Market Structure

Economists have developed four principal models of market structure: *perfect competition*, *monopoly*, *oligopoly*, and *monopolistic competition*.

This system of market structures is based on two dimensions:

- The number of producers in the market (one, few, or many)
- Whether the goods offered are identical or *differentiated*

Types of Market Structure



Market Structure Comparison

	Number of firms	Market power	Elasticity of demand	Product differentiation	Profit maximization	Pricing power
Perfect competition	Infinite	None	perfectly elastic	None	$P = MR = MC$	Price taker
Monopolistic competition	Many	Low	high elastic (in long run)	High	$MR = MC$	Price setter
Oligopoly	Few	High	relatively inelastic	Yes	$MR = MC$	Price setter
Monopoly	One	High	Relatively inelastic	Absolute	$MR = MC$	Price setter

Monopoly, Oligopoly, Monopolistic Competition

A **monopolist** is a firm that is the only producer of a good that has no close substitutes. An industry controlled by a monopolist is known as a **monopoly**.

An **oligopoly** is an industry with only a small number of producers. A producer in such an industry is known as an **oligopolist**.

When no one firm has a monopoly, but producers nonetheless realize that they can affect market prices, an industry is characterized by **imperfect competition**.

imperfect competition

Monopoly, Oligopoly, Monopolistic Competition are all imperfect competition

Monopoly

What Monopolists Do

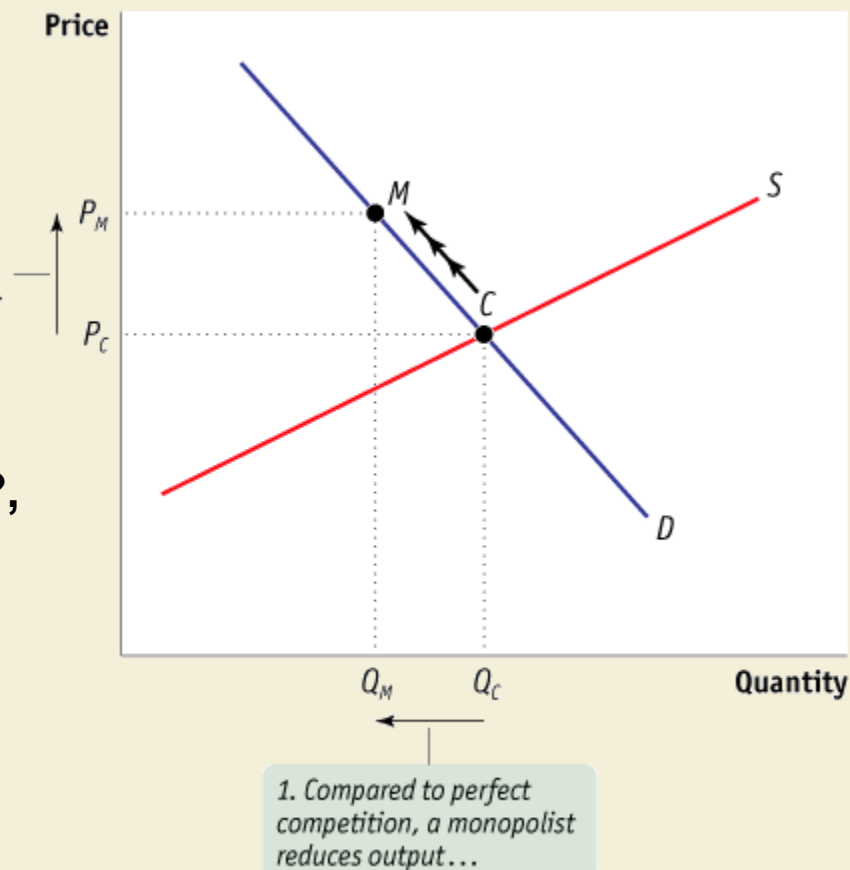
Figure 11-2

What a Monopolist Does

Under perfect competition, the price and quantity are determined by supply and demand. Here, the equilibrium is at C , where the price is P_C and the quantity is Q_C . A monopolist reduces the quantity supplied to Q_M , and moves up the demand curve from C to M , raising the price to P_M .

Can monopolist make **revenue** increase by reducing the quantity?, or can it make more profit?

We need detailed study.



The Meaning of Monopoly

What Monopolists Do

Market power is the ability of a producer to raise prices. Market power depends on the price elasticity of demand. The less elasticity, the more market power the monopolists can have.

Why Do Monopolies Exist?

To earn profits, a monopolist must be protected by a **barrier to entry**—something that prevents other firms from entering the industry.

The Meaning of Monopoly

Why Do Monopolies Exist?

Control of a Scarce Resource or Input

A monopolist that controls a resource or input crucial to an industry can prevent other firms from entering its market.

Economies of Scale

A **natural monopoly** exists when economies of scale provide a large cost advantage to having all of an industry's output produced by a single firm.

The Meaning of Monopoly

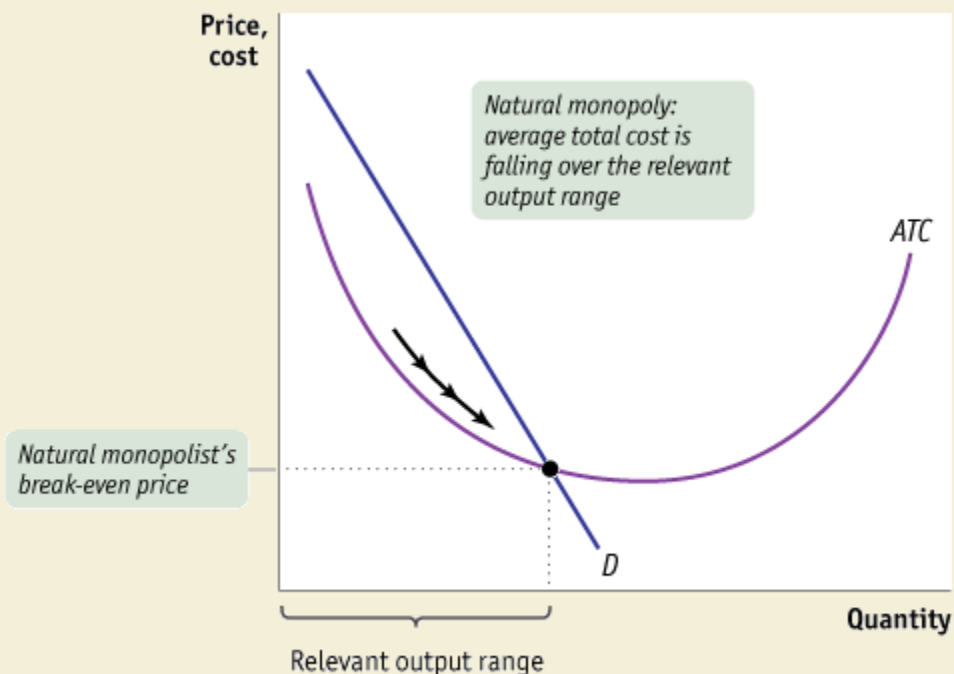
Why Do Monopolies Exist?

Economies of Scale

Figure 11-3

Economies of Scale Create Natural Monopoly

A natural monopoly can arise when fixed costs required to operate are very high. When this occurs, the firm's *ATC* curve declines over the range of output at which price is greater than or equal to average total cost. This gives the firm economies of scale over the entire range of output at which the firm would at least break even in the long run. As a result, a given quantity of output is produced more cheaply by one large firm than by two or more smaller firms.



The Meaning of Monopoly

Why Do Monopolies Exist?

Technological Superiority

A firm that maintains a consistent technological advantage over potential competitors can establish itself as a monopolist.

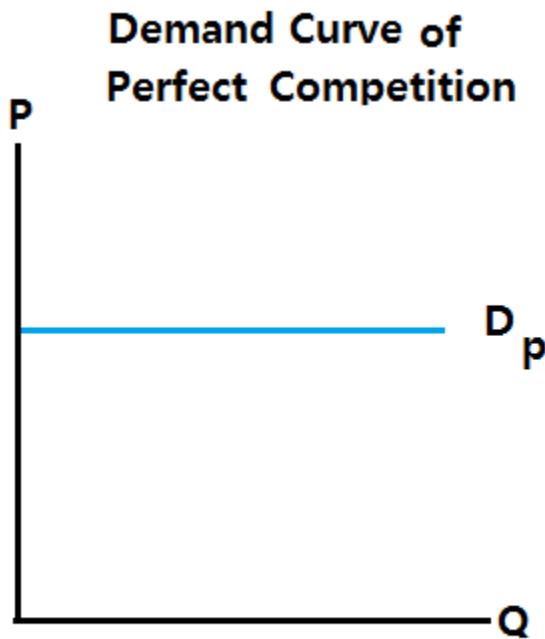
We should note, however, that in certain high-tech industries, technological superiority is not a guarantee of success against competitors.

Government-Created Barriers

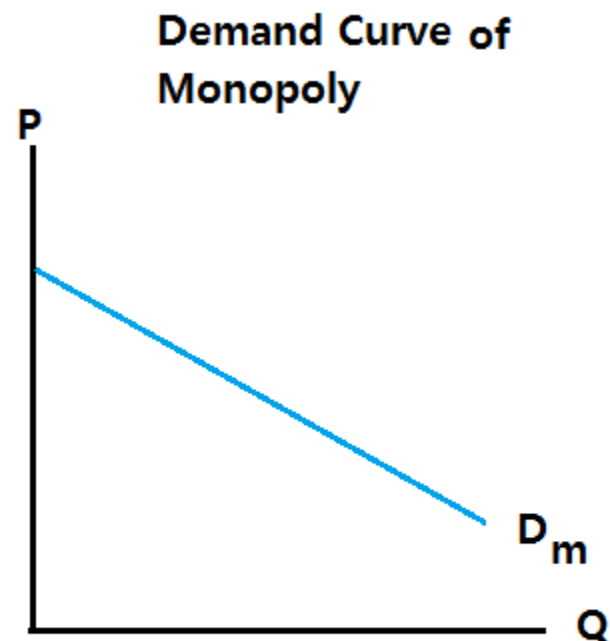
The most important legally created monopolies today arise from *patents* and *copyrights*.

Difference in Demand Curve, Marginal Revenue between Perfect Competition and Monopoly

1. Demand Curve Difference



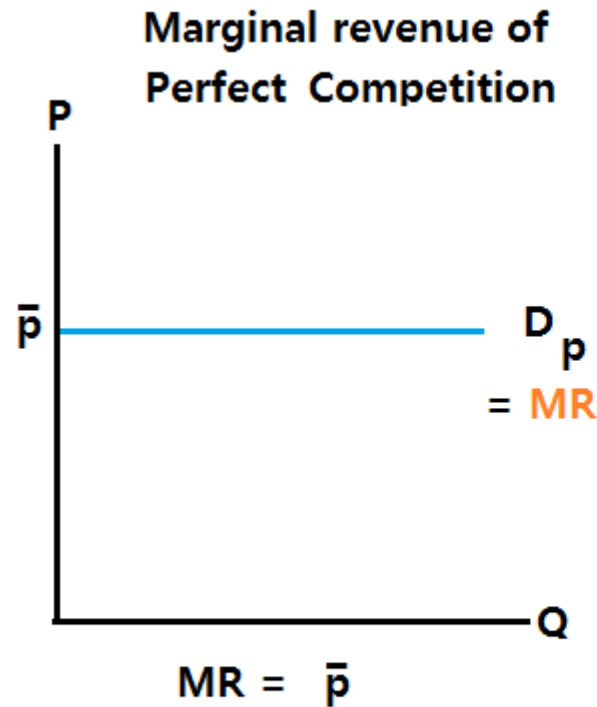
Price taker



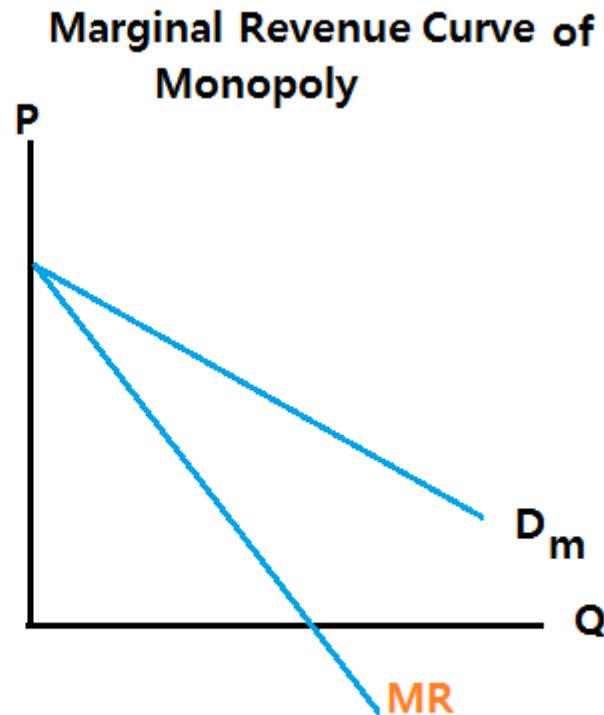
The sole supplier faces the whole industry demand curve

Difference in Demand Curve, Marginal Revenue between Perfect Competition and Monopoly

2. Marginal Revenue Difference



Marginal revenue is fixed at $\bar{p}$



Marginal revenue is decreasing and less than price

Mathematical Comparison of the Marginal Revenue

1. Perfect Competition

$$\text{Total Revenue (TR)} = TR(Q) = \bar{P} * Q$$

$$\text{Marginal Revenue (MR)} = MR(Q) = \frac{dTR(Q)}{dQ} = \bar{P}$$

2. Monopoly

$$\text{Total Revenue (TR)} = TR(Q) = P(Q) * Q$$

$$\text{Marginal Revenue (MR)} = MR(Q) = \frac{dTR(Q)}{dQ} = \frac{dP(Q)}{dQ} * Q + P(Q)$$

$$\text{where } \frac{dP(Q)}{dQ} < 0$$

Marginal Revenue and Price Elasticity of Demand

$$MR(Q) = \frac{dTR(Q)}{dQ} = \frac{dP(Q)}{dQ} Q + P(Q)$$

$$= P(Q) * \frac{\frac{dP(Q)}{P(Q)}}{\frac{dQ}{Q}} + P(Q)$$

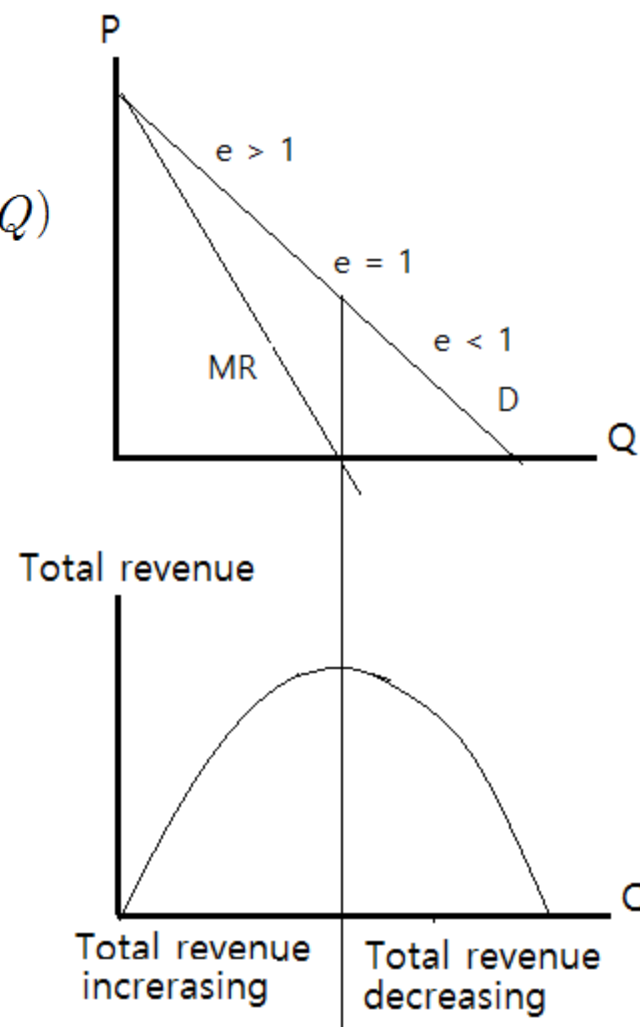
$$= P(Q) \left(\frac{1}{-e} + 1 \right)$$

$$MR(Q) > 0 \text{ If } e > 1$$

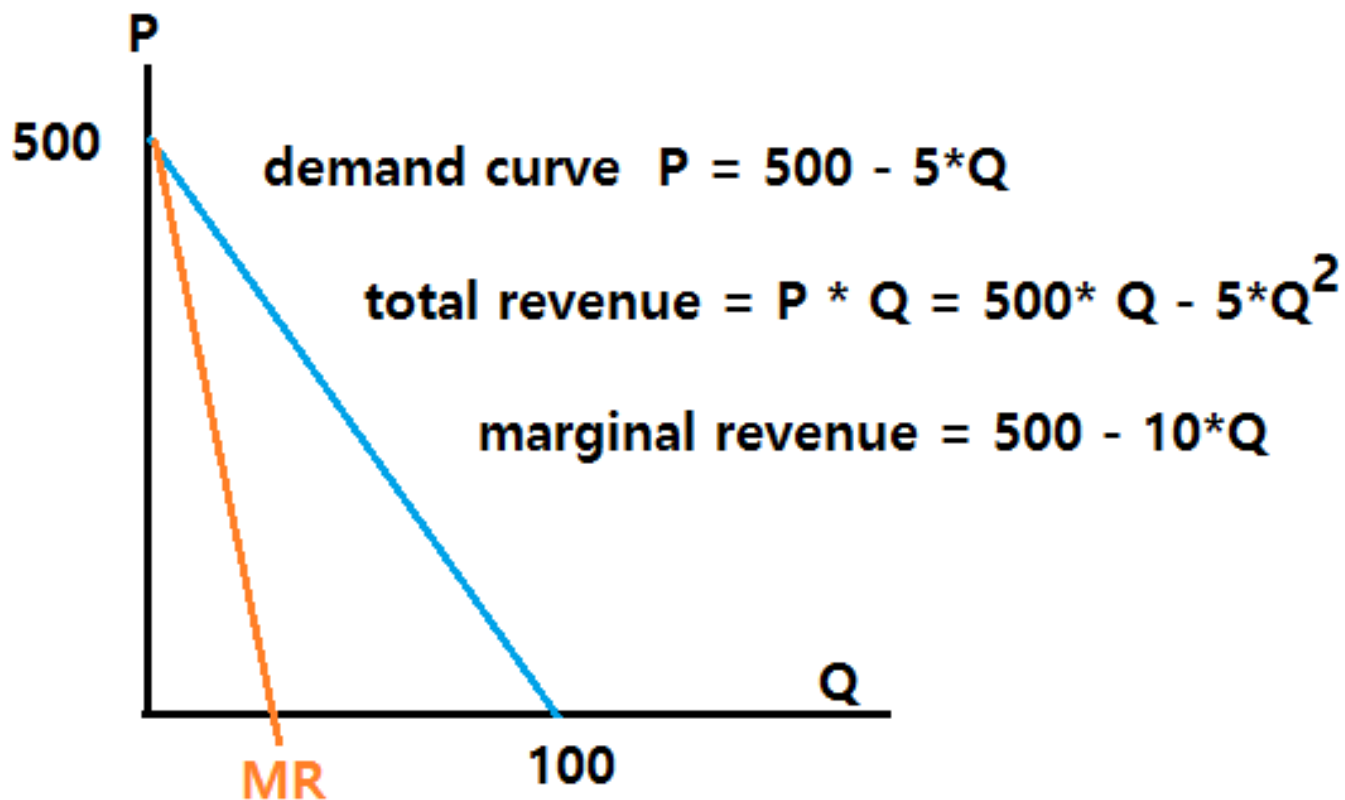
$$MR(Q) < 0 \text{ If } e < 1$$

$$MR(Q) = 0 \text{ If } e = 1$$

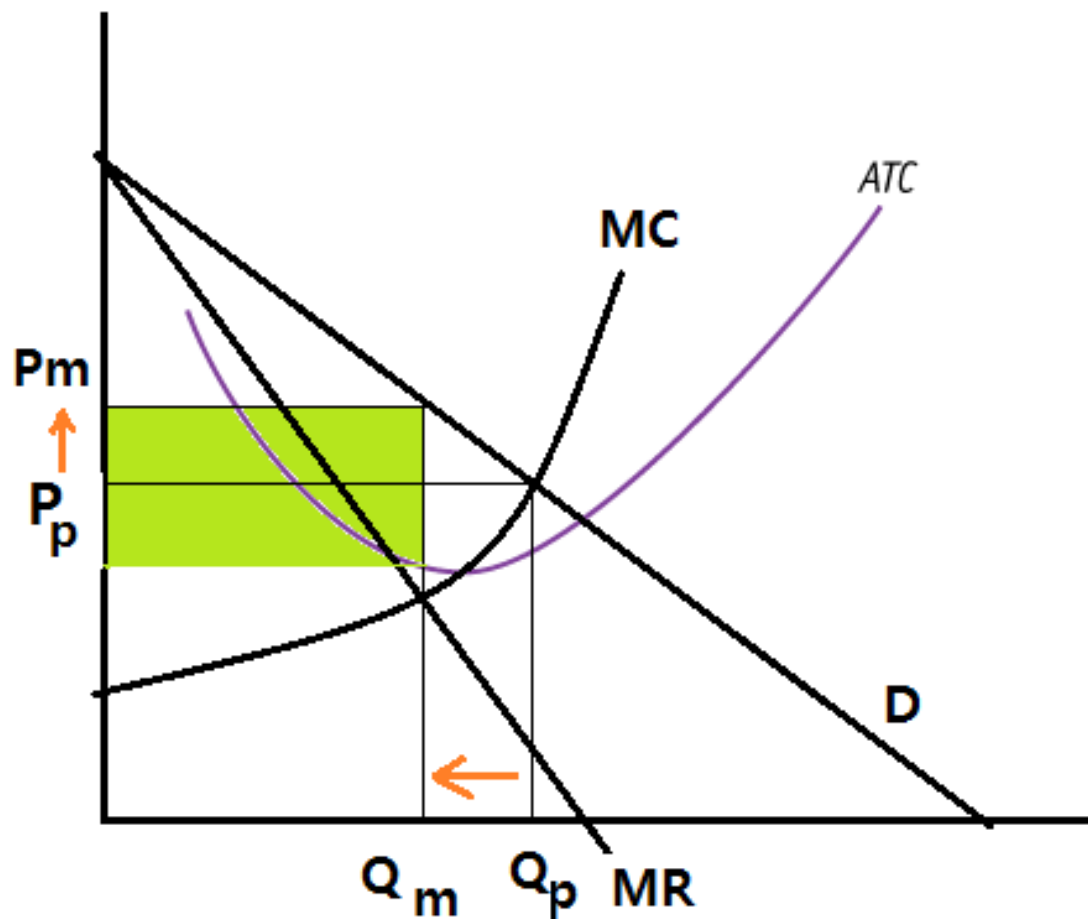
$$\text{If } e \rightarrow \infty, MR(Q) = P(Q)$$



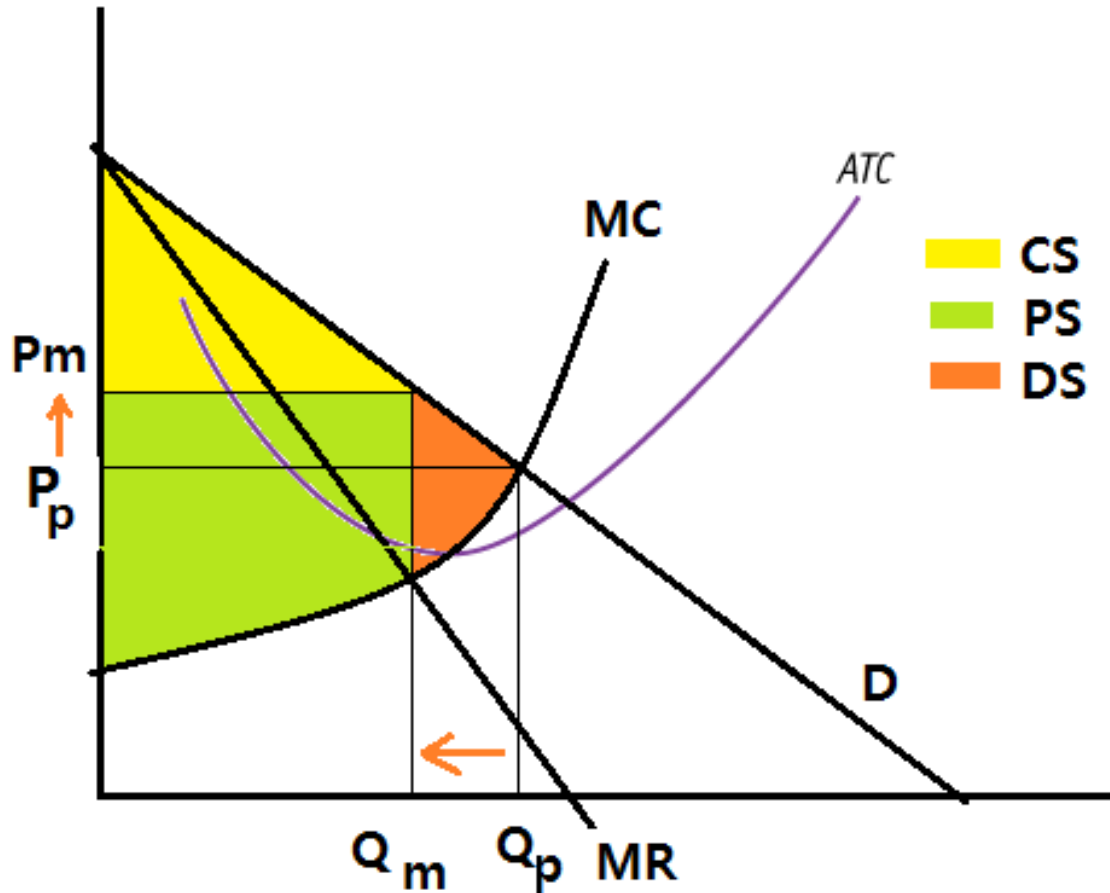
Example of marginal revenue



Profit of Monopoly



Quantity and Price for the Profit Maximizing Monopoly



How a Monopolist Maximizes Profit

The Monopolist's Profit-Maximizing Output and Price

The monopolist maximizes its profit by using the optimal output rule:

$MR = MC$ at the monopolist's profit-maximizing quantity of output

Monopoly versus Perfect Competition

$P = MC$ at the perfectly competitive firm's profit-maximizing quantity of output

$P > MR = MC$ at the monopolist's profit-maximizing quantity of output

Monopoly and Public Policy

Preventing Monopoly

Policy toward monopoly depends crucially on whether or not the industry in question is a natural monopoly, one in which economies of scale ensure that bigger producers have lower average total cost.

Breaking up a monopoly that isn't natural is clearly a good idea: the gains to consumers outweigh the loss to the producer.

It is not so clear whether a natural monopoly should be broken up, because this would raise average total cost.

Yet even in the case of a natural monopoly, a profit-maximizing monopolist acts in a way that causes inefficiency.

Monopoly and Public Policy

Dealing with Natural Monopoly

(1) Public Ownership

In **public ownership** of a monopoly, the good is supplied by the government or by a firm owned by the government.

(2) Regulation

Price regulation limits the price that a monopolist is allowed to charge.

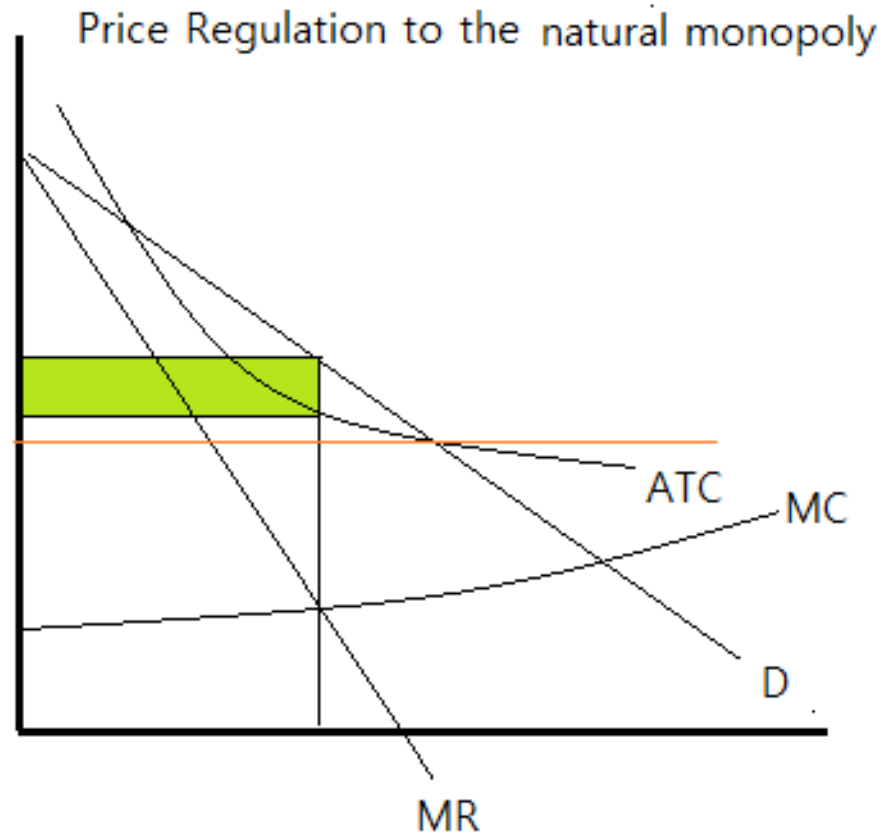
(3) Keep competition

Anti-trust law prohibits M&A if it becomes monopoly
Sometimes governments order to breaking-up monopoly firm

Monopoly and Public Policy

Dealing with Natural Monopoly : Average total cost is decreasing

Price Regulation



Price Discrimination

A **single-price monopolist** offers its product to all consumers at the same price. Sellers engage in **price discrimination** when they charge different prices to different consumers.

Examples:	Low price	High price
	Children	Parents
	Senior citizens	non-senior citizens
	Hard-cover books	Paper books
	Theater movies	Video
	Morning movies	Regular-time movies
	Student price	Business class

The lower price group has higher elasticity

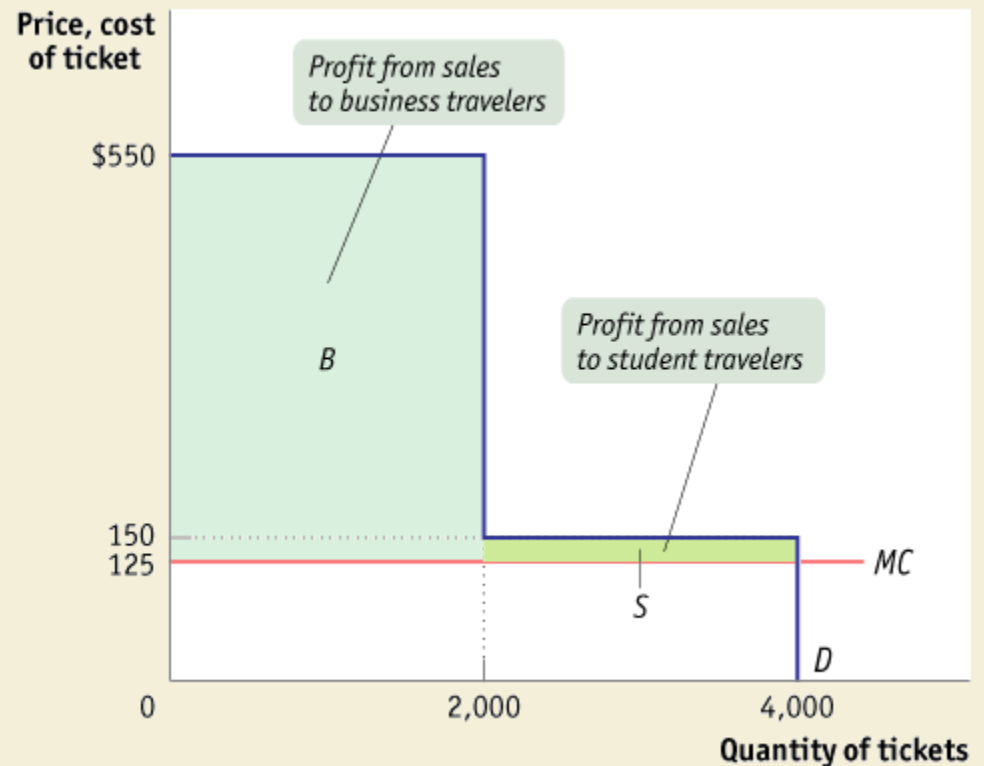
Price Discrimination

The Logic of Price Discrimination

Figure 11-10

Two Types of Airline Customers

Air Sunshine has two types of customers, business travelers willing to pay \$550 per ticket and students willing to pay \$150 per ticket. There are 2,000 of each kind of customer. Air Sunshine has constant marginal cost of \$125 per seat. If Air Sunshine could charge these two types of customers different prices, it would maximize its profit by charging business travelers \$550 and students \$150 per ticket. It would capture all of the consumer surplus as profit.



Price Discrimination

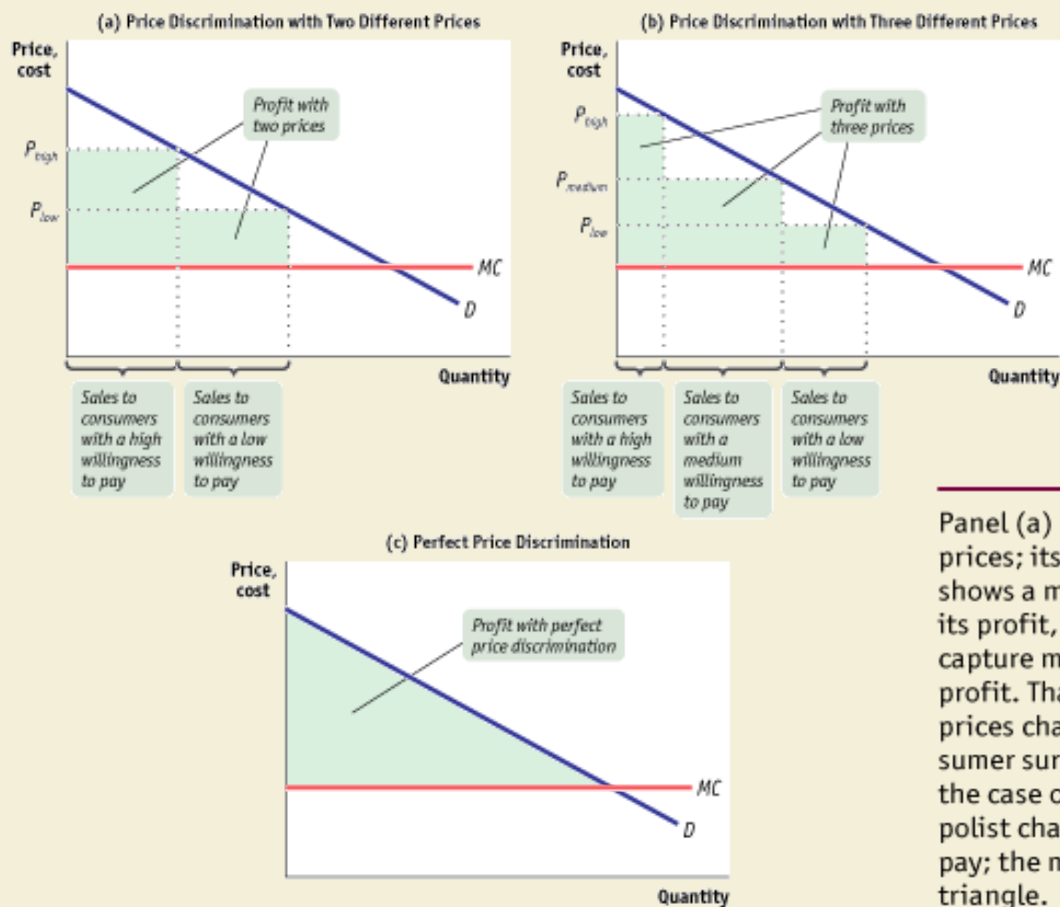
For the price discrimination to be possible, markets must be separated each other (price discrimination cannot be hold if you buy from one market and resell it at another market). Also the demand elasticities must be different.

Perfect price discrimination takes place when a monopolist charges each consumer his or her willingness to pay—the maximum that consumer is willing to pay.

Price Discrimination

Perfect Price Discrimination

Figure 11-11 Price Discrimination



Panel (a) shows a monopolist that charges two different prices; its profit is shown by the shaded area. Panel (b) shows a monopolist that charges three different prices; its profit, too, is shown by the shaded area. It is able to capture more of the consumer surplus and to increase its profit. That is, by increasing the number of different prices charged, the monopolist captures more of the consumer surplus and makes a large profit. Panel (c) shows the case of perfect price discrimination, where a monopolist charges each consumer his or her willingness to pay; the monopolist's profit is given by the shaded triangle.

Kinds of Price Discrimination

- *First Degree Price Discrimination:* The monopoly seller knows the absolute maximum price that every consumer is willing to pay
- *Second Degree Price Discrimination:* Price varies according to quantity demanded. Larger quantities are available at a lower unit price
- *Third Degree Price Discrimination:* Price differs to different consumer groups

KEY TERMS

Monopolist

Monopoly

Market power

Barrier to entry

Natural monopoly

Public ownership

Price regulation

Single-price monopolist

Price discrimination

Perfect price discrimination